

CHAPTER – 5

ORDINARY LINEAR DIFFERENTIAL EQUATIONS OF SECOND AND HIGHER ORDER

5.1 General

As discussed earlier, a **linear differential equation** is the one which is of first degree in the dependent variable (usually y) and its derivatives, does not contain products of the dependent variable and/or its derivatives, and also does not have transcendental functions of the dependent variable. For example,

$$\begin{array}{ll} y''' + x^2 y'' + \ln x y' + x^8 y = 0 & \text{is linear,} \\ x^2 y' + 2y = 0 & \text{is linear} \\ y'' + y y' = 0 & \text{is nonlinear} \\ 3 (y')^2 + y = 0 & \text{is nonlinear} \end{array}$$

In this chapter we shall discuss linear differential equations, particularly those having constant coefficients. These equations represent or apply to many interesting and important natural phenomena. As such they are of great significance in physics and engineering, especially in connection with vibrations in mechanics and the theory of electric circuits. Moreover, many profound and beautiful ideas in pure mathematics have grown out of the study of these equations.

The general form of the **linear differential equation of n th order** is,

$$P_n \frac{d^n y}{dx^n} + P_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + P_2 \frac{d^2 y}{dx^2} + P_1 \frac{dy}{dx} + P_0 y = Q \quad (5.1.1)$$

where, P 's and Q 's are constants or functions of x . No method has been developed to solve all the linear differential equations, and the solvable cases are obtained by specializing the coefficients P_i and the right-hand member Q as given below,

- a. If the functions P_i are all constants, Eqn.(5.1.1) is said to be **linear differential equation with constant coefficients**.

- b. If $Q = 0$, Eqn.(5.1.1) is called a **homogeneous linear differential equation** (the word homogeneous used here has a very different meaning from that used in sec. 3.3 for the first order differential equations). It is also called a **reduced linear differential equation**, and is given as,

$$p_n \frac{d^n y}{dx^n} + p_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + p_2 \frac{d^2 y}{dx^2} + p_1 \frac{dy}{dx} + p_0 y = 0. \quad (5.1.2)$$

- c. If $Q \neq 0$, Eqn.(5.1.1) is called a **non-homogeneous** or **complete linear differential equation**.

5.2 Search for the Method for Solution of Second Order Differential Equations

It may be recalled that we can solve a first-order, non-exact linear differential equation, by multiplying it with an integrating factor. One may like to explore the possibility of solving the second order linear differential equations by the method of integrating factor as well. In fact this idea also came to the mind of the renowned mathematician **Joseph Louis Lagrange (1736-1813)**. He proved the existence of such an integrating factor under certain conditions over 200 years ago. However, he could not find the same. In the process of finding such an integrating factor, he proved that the given differential equation would give rise to another differential equation that is equally difficult to solve as the original equation. Actually **Lagrange** tried to find an integrating factor $\mu(x)$, so that on multiplying it to the linear differential equation,

$$y'' + p(x) y' + q(x) y = f(x), \quad (5.2.1)$$

it would yield an equation of the form,

$$\frac{d}{dx} [\mu(x) y' + g(x) y] = \mu(x) f(x), \quad (5.2.2)$$

where $g(x)$ is a function that could be determined from $p(x)$ and $q(x)$. Accordingly, integration of this equation would yield the first-order differential equation,

$$\mu(x) y' + g(x) y = \int \mu(x) f(t) dt, \quad (5.2.3)$$

which could be solved by some method for the solution of first-order linear differential equations. However, this method is not that simple as it involves evaluation of $\mu(x)$ before solving Eqn.(5.2.3). In order to find the value of $\mu(x)$, we would put the value of $f(x)$ from Eqn.(5.2.1) into Eqn.(5.2.2) to get,

$$\frac{d}{dx} [\mu(x) y' + g(x) y] = \mu(x) [y'' + p(x) y' + q(x) y].$$

On comparing the coefficients of y' and y , this equation gives,

$$\mu(x) p(x) = \mu'(x) + g(x), \quad (5.2.4)$$

$$\mu(x) q(x) = g'(x). \quad (5.2.5)$$

On differentiating Eqn.(5.2.4) and substituting the value of $g'(x)$ from Eqn (5.2.5) into it we obtain,

$$\mu''(x) - p(x) \mu'(x) + [q(x) - p'(x)] \mu(x) = 0. \quad (5.2.6)$$

This is called **adjoint equation**, which is as hard to solve as the original equation. Thus **Lagrange's** method of integrating-factor amounts to replacing a differential equation by another equally difficult differential equation. Therefore, this method is not helpful in solving the second-order differential equations. It may be seen that Lagrange's integrating-factor method works well in solving the first-order differential equation,

$$y' + p(x) y = q(x), \quad (5.2.7)$$

because the **adjoint equation** turns out to be,

$$\mu'(x) - p(x) \mu(x) = 0, \quad (5.2.8)$$

which can be solved by the method of **separation of variables**, and it gives,

$$\mu(x) = e^{\int p(x) dx}. \quad (5.2.9)$$

After being unsuccessful in employing the method of integrating-factor for second-order differential equations we continue our search for the general solution of these equations. For that we begin with one of the most important properties of linear homogeneous differential equations, discussed below.

5.3 Principle of Superposition for Homogeneous Linear Differential Equations

The homogeneous linear differential equations have the following very important properties,

- a. The sum of any two or more solutions of an equation is also a solution of this equation.
- b. A constant multiple of a solution of an equation is also a solution of this equation.

These properties form the so called **Principle of Superposition** for homogeneous linear differential equations, as discussed below.

Theorem 5.3.1 Principle of Superposition for Homogeneous Linear Differential Equations

If y_1 and y_2 are any two solutions of the homogeneous linear differential equation,

$$y'' + p(x) y' + q(x) y = 0, \quad (5.3.1)$$

on some interval I, then any linear combination,

$$y(x) = c_1 y_1(x) + c_2 y_2(x), \quad (5.3.2)$$

where c_1 and c_2 are arbitrary constants, is also a solution of equation (5.3.1) on the same interval I .

Proof

We substitute the value of y from Eqn (5.3.2) and its derivatives into Eqn (5.3.1) to get,

$$\begin{aligned} y'' + p(x) y' + q(x) y &= [c_1 y_1 + c_2 y_2]'' + p(x) [c_1 y_1 + c_2 y_2]' + q(x) [c_1 y_1 + c_2 y_2] \\ &= c_1 y_1'' + c_2 y_2'' + c_1 p(x) y_1' + c_2 p(x) y_2' + c_1 q(x) y_1 + c_2 q(x) y_2 \\ &= c_1 [y_1'' + p(x) y_1' + q(x) y_1] + c_2 [y_2'' + p(x) y_2' + q(x) y_2] \\ &= c_1 [0] + c_2 [0] \\ &= 0, \end{aligned}$$

because both y_1 and y_2 were assumed to be the solutions of Eqn (5.3.1). Therefore, regardless the values of c_1 and c_2 , the value of y as given by Eqn (5.3.2) satisfies the differential equation (5.3.1). This proves that Eqn (5.3.2) is a solution of differential equation (5.3.1).

As a particular consequence of the above result, we note that any constant multiple of a solution of the differential equation (5.3.1) is also a solution of this differential equation.

Example 5.3.1

Verify that $y_1(x) = x^2$, and $y_2(x) = x^{-1}$ are two solutions of the differential equation $x^2 y'' - 2y = 0$, $x > 0$. Then show that, $y = c_1 x^2 + c_2 x^{-1}$ is also a solution of this equation for any c_1 and c_2 .

Solution

We see that differentiating twice, $y_1 = x^2$ gives, $y_1'' = 2$. Therefore, the given differential equation, $x^2 y'' - 2y = 0$, yields,

$$x^2 y_1'' - 2y_1 = 2x^2 - 2x^2 = 0.$$

Thus the solution $y_1(x) = x^2$ satisfies the given differential equation.

Also, for $y_2(x) = x^{-1}$, we obtain $y_2'' = 2x^{-3}$. Substituting these values in the given differential equation we get,

$$x^2 y_2'' - 2y_2 = x^2 (2x^{-3}) - 2x^{-1} = 0.$$

Accordingly, we see that $y_2(x) = x^{-1}$ also satisfies the given differential equation.

Next we shall show that the linear combination of these two solutions, i.e., $y = c_1 x^2 + c_2 x^{-1}$ also satisfies this differential equation. Differentiating it twice we obtain, $y'' = c_1 (2) + c_2 (2x^{-3})$. Therefore, substituting these values in the given differential equation we obtain,

$$x^2 y'' - 2y = x^2 [2c_1 + 2c_2 x^{-3}] - 2 [c_1 x^2 + c_2 x^{-1}] = 0.$$

Therefore, $y = c_1 x^2 + c_2 x^{-1}$ also satisfies the given differential equation.

Hence, it is proved that $y_1 = x^2$, $y_2 = x^{-1}$, and their linear combination, $y = c_1 x^2 + c_2 x^{-1}$ are all solutions of the given differential equation.

Having discussed the *Principle of Superposition for Homogeneous Differential Equations*, next we discuss the *Initial Value Problem for Second-Order Differential Equations*.

Initial Value Problem for Second-Order Differential Equations

We know that the general solution of the **first-order differential equation** contains **one arbitrary constant**. Furthermore, we know from calculus that the general solution of the trivial second-order differential equation $y'' = 0$ is $y(x) = c_1 + c_2 x$. This shows that **the number of arbitrary constants in the solution of a differential equation equals its order**. Correspondingly, as many conditions are required to find the values of these arbitrary constants. These views can be summarized in the form of *initial value problem for second-order differential equation*, as stated below:

A problem of finding the solution of the differential equation,

$$F(x, y, y', y'') = 0, \quad (5.3.3)$$

where,

$$y(x_0) = y_0, \text{ and } y'(x_0) = y_1, \quad (5.3.4)$$

on an interval I , is called an *initial value problem* for a second-order differential equation, where, y_0 and y_1 are given constants.

Example 5.3.2

Given that $y_1(x) = e^{2x} \cos 3x$, and $y_2(x) = e^{2x} \sin 3x$, are solutions of the linear differential equation,

$$y'' - 4y' + 13y = 0. \quad (5.3.5)$$

Find a solution to this equation satisfying the initial conditions,

$$y(0) = 5, \text{ and } y'(0) = -2. \quad (5.3.6)$$

Solution

According to the *principle of superposition for homogeneous linear differential equations*, any linear combination of the form,

$$y = c_1 e^{2x} \cos 3x + c_2 e^{2x} \sin 3x, \quad (5.3.7)$$

with c_1 and c_2 as arbitrary constants is a solution to Eqn (5.3.5). We shall try to find the values of c_1 and c_2 so as to satisfy the initial conditions (5.3.6).

On differentiating Eqn (5.3.7) we obtain,

$$y' = c_1 (2 e^{2x} \cos 3x - 3 e^{2x} \sin 3x) + c_2 (2 e^{2x} \sin 3x + 3 e^{2x} \cos 3x). \quad (5.3.8)$$

Using the initial conditions (5.3.6), Eqns (5.3.7) and (5.3.8) yield $c_1 = 5$, and $c_2 = -4$. Thus the solution to Eqn (5.3.5) that satisfies the initial conditions (5.3.6) is given as,

$$y = 5 e^{2x} \cos 3x - 4 e^{2x} \sin 3x$$

5.4 Existence and Uniqueness of a Solution for an Initial Value Problem

Before actually solving the second-order linear differential equation it would be appropriate to know if and when a solution indeed exists, and the interval on which it exists. Further, it would be worthwhile to know the number of solutions. However, it may be recalled that all types of linear equations including linear differential equations, linear difference equations, linear algebraic equations, and even the systems of linear equations have a common property that the **number of solutions** is always **zero**, **one**, or **infinity**. There is no possibility of obtaining *two, three, four, ...,* solutions. The following *Existence and Uniqueness Theorem* provides conditions **sufficient** for the second-order initial value problem to have exactly **one** or **unique** solution.

Theorem 5.4.1 Existence and Uniqueness Theorem

Suppose $p(x)$, $q(x)$, and $f(x)$ are continuous functions on a open interval I that contains a point x_0 . Then for any choice of the initial values y_0 and y_1 , there exists **one and only one** (i.e., **unique**) solution $y(x)$ on the interval I to the initial value problem,

$$y'' + p(x) y' + q(x) y = f(x), \quad (5.4.1)$$

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad (5.4.2)$$

The proof of this theorem is quite cumbersome and is available in the literature.

Note:

Geometrically, the uniqueness part of the above theorem asserts that two solution curves passing through the same point and with the same slope must be identical throughout the interval I .

Example 5.4.1 (Verifying the Existence and Uniqueness Theorem)

Determine the largest interval that contains $x = 0$, on which a unique solution for the following initial value problem exists:

$$y'' + \frac{y}{x-1} = e^{-x}, \quad (5.4.3)$$

$$y(0) = 1, \quad y'(0) = 0. \quad (5.4.4)$$

Solution.

Comparing the given differential equation (5.4.3) with the standard second-order linear differential equation (5.4.1) we have,

$$p(x) = 0, q(x) = \frac{1}{x-1}, \text{ and } f(x) = e^{-x}.$$

Thus, $x = 1$ is the only value of x for which all these functions are NOT continuous, as $q(x)$ is not defined at this point. However, the initial conditions (5.4.4) are defined at the initial point $x_0 = 0$. Therefore, the *Existence and Uniqueness Theorem* guarantees that a unique solution would exist on the interval $(-\infty, 1)$. We may not know the solution itself, however, we do know that one does exist definitely. Moreover, we also know that a solution does not exist on any interval that contains $x = 1$.

EXERCISE-5.1

(Existence and Uniqueness of Solutions)

Note: Answers are given against the respective questions within square brackets.

1. Verify that e^{-x} and e^{2x} and any linear combination $c_1 e^{-x} + c_2 e^{2x}$ are all solutions of the differential equation,

$$y'' - y' - 2y = 0$$
2. Verify that e^{-x} and e^x and any linear combination $c_1 e^{-x} + c_2 e^x$ are all solutions of the differential equation,

$$y'' - y = 0.$$
3. Show that the circular functions, $\sin x$ and $\cos x$, are solutions of the differential equation $y'' + y = 0$, whereas, the hyperbolic functions, $\sinh x$ and $\cosh x$, are solutions of $y'' - y = 0$.
4. If every solution of the differential equation, $y'' + y = 0$, has the form $y = c_1 \cos x + c_2 \sin x$, where c_1 and c_2 are arbitrary constants, then verify the following:
 - a. There is a **unique solution** to the given differential equation satisfying the boundary conditions: $y(0) = 1$, and $y(\pi/2) = 0$
 - b. There is **no solution** to the given differential equation that satisfies the boundary conditions: $y(0) = 1$, and $y(\pi) = 0$.
 - c. There are **infinite number of solutions** to the given differential equation satisfying the boundary conditions: $y(0) = 1$, and $y(\pi) = -1$.
5. For the following problems, use the *Existence and Uniqueness Theorem* to determine intervals for which, certainly there exists, a unique solution to the given differential equation with the initial conditions: $y(x_0) = y_0$, and $y'(x_0) = y'_0$:
 - a. $y'' + 3y = 0$ $[(-\infty, \infty) \text{ for any } x_0]$
 - b. $y'' + 2y' + y = 1$ $[(-\infty, \infty) \text{ for any } x_0]$
 - c. $y'' + xy' + 3y = \sin x$ $[(-\infty, \infty) \text{ for any } x_0]$
 - d. $xy'' + 5y = 1$ $[(-\infty, 0) \text{ if } x_0 < 0, (0, \infty) \text{ if } x_0 > 0]$
 - e. $x(x-2)y'' + 3xy' + y = 0$ $[(-\infty, 0) \text{ if } x_0 < 0, (0, 2) \text{ if } 0 < x_0 < 2, (2, \infty) \text{ if } x_0 > 2]$
 - f. $y'' + (\sin x)y' + 2y = 4$ $[(-\infty, \infty) \text{ for any } x_0]$
 - g. $y'' + x(3-x)y' + 3y = 4$ $[(-\infty, \infty) \text{ for any } x_0]$

5.5 The Wronskian and Linearly Independent Solutions

The **Wronskian** is named for the mathematician **Jozef Maria Hoene-Wronski (1778 - 1853)**, who was born in Poland and educated in Germany. He lived most of his life in France. More a Philosopher than a mathematician he believed that absolute truth could be attained only through mathematical reasoning. His only noteworthy contribution to mathematics was the Wronskian determinant. Always an eccentric, he would indulge in heated disputes with individuals and institutions. Eventually he went insane.

Definition of Wronskian

Given n differentiable functions $f_1(x)$, $f_2(x)$, $f_3(x)$, ..., $f_n(x)$, possessing at least $n-1$ derivatives, the determinant,

$$W = \begin{vmatrix} f_1 & f_2 & f_3 & \dots & f_n \\ f_1' & f_2' & f_3' & \dots & f_n' \\ f_1'' & f_2'' & f_3'' & \dots & f_n'' \\ \dots & \dots & \dots & \dots & \dots \\ f_1^{(n-1)} & f_2^{(n-1)} & f_3^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix},$$

is called the Wronskian of the functions $f_1, f_2, f_3, \dots, f_n$.

Theorem 5.5.1 (Criterion for Linearly Independent Functions/Solutions)

Suppose the functions (or solutions): $f_1(x)$, $f_2(x)$, $f_3(x)$, ..., $f_n(x)$, possess at least $n-1$ derivatives. Then,

- a. **Linearly Independent Functions.** These functions are **linearly independent** on an interval I , if their **Wronskian** $W \neq 0$, for at least one point on this interval.
- a. **Linearly Dependent Functions.** These functions are **linearly dependent** on an interval I , if their **Wronskian** $W = 0$, for every $x \in I$.

Example 5.5.1

Determine whether the functions $f_1(x) = \sin^2 x$ and, $f_2(x) = 1 - \cos 2x$, are linearly dependent or not on $(-\infty, \infty)$.

Solution

The Wronskian of the given functions is:

$$W = \begin{vmatrix} \sin^2 x & 1 - \cos 2x \\ 2 \sin x \cos x & 2 \sin 2x \end{vmatrix} = \begin{vmatrix} \sin^2 x & 2 \sin^2 x \\ \sin 2x & 2 \sin 2x \end{vmatrix} = 2 \sin^2 x \sin 2x - 2 \sin^2 x \sin 2x = 0.$$

This shows that the **given functions are linearly dependent**

5.6 Abel's Formula/Theorem

If y_1, y_2 , are the solutions of the differential equation,

$$y'' + p(x) y' + q(x) y = 0, \quad (5.6.1)$$

where p and q are continuous on an open interval I , then the Wronskian $W(y_1(x), y_2(x))$ is given by,

$$W = c e^{-\int p dx}, \quad (5.6.2)$$

where c is a certain constant that depends on y_1 and y_2 , but not on x .

Furthermore, **W is either zero** for all $x \in I$ (if $c = 0$) **or else, never zero** on I (if $c \neq 0$).

Proof

Since y_1 and y_2 are solutions of differential equation (5.6.1), so they satisfy it and we have

$$y_1'' + p(x) y_1' + q(x) y_1 = 0$$

$$y_2'' + p(x) y_2' + q(x) y_2 = 0$$

Multiplying the 1st equation by $-y_2$, the 2nd by y_1 , and adding the resulting equations, we obtain,

$$(y_1 y_2'' - y_1'' y_2) + p (y_1 y_2' - y_1' y_2) = 0. \quad (5.6.3)$$

Here, the Wronskian $W(x) = W(y_1(x), y_2(x))$ is given as,

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2. \quad (5.6.4)$$

Differentiating it we obtain,

$$\begin{aligned} W'(x) &= (y_1 y_2'' + y_1' y_2') - (y_1' y_2' + y_1'' y_2) \\ &= y_1 y_2'' - y_1'' y_2. \end{aligned} \quad (5.6.5)$$

Substituting the values of $W(x)$ and $W'(x)$ from Eqns. (5.6.4) and (5.6.5) into Eqn. (5.6.3) we obtain,

$$W'(x) + p W(x) = 0. \quad (5.6.6)$$

On solving this 1st order linear differential equation of **separation of variables** form, we get,

$$W(x) = c e^{-\int p dx}, \quad (5.6.7)$$

where c is a constant of integration.

Notes

- a. We see that, since exponential function is never zero, so, $W \neq 0$, unless $c = 0$, in which case $W = 0$ for all x .
- b. **The Wronskian W of any two fundamental sets of solutions of the same differential equation can differ only by a multiplicative constant, and that the Wronskian of any fundamental set of solutions can be determined up to a multiplicative constant, without solving the differential equation.**

5.7 Fundamental Set of Solutions

The set of n linearly independent solutions, $y_1(x)$, $y_2(x)$, $y_3(x)$, ..., $y_n(x)$, of an n th order homogeneous linear differential equation,

$$p_0(x) y^{(n)} + p_1(x) y^{(n-1)} + p_2(x) y^{(n-2)} + \dots + p_n(x) y = 0 \quad (5.7.1)$$

is called a **Fundamental Set of Solutions** of this equation. Moreover, the **linear combination** of the above set of solutions,

$$y = c_1 y_1 + c_2 y_2 + c_3 y_3 + \dots + c_n y_n$$

is called the **General Solution** of Eqn. (5.7.1).

Example 5.7.1

Show that the two solutions, $y_1(x) = e^x$, and $y_2(x) = e^{-2x}$, constitute a fundamental set of solutions of the differential equation,

$$y'' + y' - 2y = 0, \quad (5.7.2)$$

and find the general solution.

Solution

We see that for $y_1(x) = e^x$, on differentiation we obtain,

$$y_1'(x) = e^x, \text{ and } y_1''(x) = e^x,$$

which satisfy Eqn.(5.7.2). Similarly, for $y_2(x) = e^{-2x}$, we obtain,

$$y_2'(x) = -2e^{-2x}, \text{ and } y_2''(x) = 4e^{-2x},$$

which also satisfy Eqn. (5.7.2). Next we evaluate the Wronskian, $W(e^x, e^{-2x})$, which is,

$$W(e^x, e^{-2x}) = \begin{vmatrix} e^x & e^{-2x} \\ e^x & -2e^{-2x} \end{vmatrix} = -2e^{-x} - e^{-x} = -3e^{-x} \neq 0. \quad (5.7.3)$$

We see that the Wronskian is non-zero on the interval $(-\infty, \infty)$. Therefore, we conclude that the two solutions, $y_1(x) = e^x$, and $y_2(x) = e^{-2x}$ are linearly independent on this interval. Hence they

constitute a *fundamental set of solutions* of the given differential equation. Accordingly the general solution of the given differential equation (5.7.2) is,

$$y = c_1 e^x + c_2 e^{-2x}, \quad (5.7.4)$$

where, c_1 , and c_2 are arbitrary constants.

EXERCISE-5.2

(Linearly Independent/Dependent Solutions)

Note: Answers are given against the respective questions within square brackets.

In each of the following problems, determine the Wronskian of the given pair of functions:

1. $y_1 = e^{2x}$, $y_2 = e^{-3x/2}$; $[-\frac{7}{2} e^{x/2}]$
2. $y_1 = e^{mx}$, $y_2 = e^{nx}$, when $m \neq n$; $[(n - m) e^{(m+n)x}]$
3. $y_1 = \cos x$, $y_2 = \sin x$; $[1]$
4. $y_1 = e^{-2x}$, $y_2 = xe^{-2x}$; $[e^{-4x}]$
5. $y_1 = x$, $y_2 = xe^x$; $[x^2 e^x]$
6. $y_1 = \sin^2 x + \cos^2 x$, $y_2 = 1$ $[0]$
7. $y_1 = e^x \sin x$, $y_2 = e^x \cos x$; $[-e^{2x}]$
8. $y_1 = \cos^2 x$, $y_2 = 1 + \cos 2x$; $[0]$
9. $y_1 = x$, $y_2 = |x|$; $[0]$
10. $y_1 = \ln x$, $y_2 = \ln x^2$; $[0]$
11. $y_1 = e^x$, $y_2 = e^{2x}$, $y_3 = e^{3x}$; $[2 e^{6x}]$

Determine whether the functions given in the following problems are linearly dependent or independent on the interval $(-\infty, \infty)$:

12. $y_1 = x$, $y_2 = x^2$, $y_3 = 4x - 3x^2$; **[Dependent]**
13. $y_1 = 5$, $y_2 = \cos^2 x$, $y_3 = \sin^2 x$; **[Dependent]**
14. $y_1 = x$, $y_2 = -1$, $y_3 = x + 3$; **[Dependent]**
15. $y_1 = 1 + x$, $y_2 = x$, $y_3 = x^2$; **[Independent]**

For the under-mentioned problems, undertake the following steps,

- (i) Verify that $y_1(x)$, and $y_2(x)$ are the solutions of the given differential equations.
- (ii) Verify that $y_1(x)$, and $y_2(x)$ are linearly independent solutions.
- (iii) Find the general solution of the given differential equation.
- (iv) Solve the given initial value problem.

16. $y'' - 5y' + 6y = 0$; $y_1(x) = e^{2x}$, $y_2(x) = e^{3x}$, $y(0) = 0$, $y'(0) = 0$;

[(iii). $y = c_1 e^{2x} + c_2 e^{3x}$; (iv). $y = 0$]

17. $y'' + 5y' + 6y = 0$; $y_1(x) = e^{-2x}$, $y_2(x) = e^{-3x}$, $y(0) = 0$, $y'(0) = 1$;

[(iii). $y = c_1 e^{-2x} + c_2 e^{-3x}$; (iv). $y = e^{-2x} + e^{-3x}$]

$$18. y'' - 5y = 0; y_1(x) = e^{\sqrt{5}x}, y_2(x) = e^{-\sqrt{5}x}; y(0) = 1, y'(0) = 0;$$

$$[(\text{iii}). y = c_1 e^{\sqrt{5}x} + c_2 e^{-\sqrt{5}x}, (\text{iv}). y = (e^{\sqrt{5}x} + c_2 e^{-\sqrt{5}x})/2]$$

$$19. y'' + 4y = 0; y_1(x) = \sin 2x, y_2(x) = \cos 2x; y(0) = 1, y'(0) = 1;$$

$$[(\text{iii}). y = c_1 \sin 2x + c_2 \cos 2x, (\text{iv}). y = (\sin 2x + 2 \cos 2x)/2]$$

$$20. y'' - y = 0; y_1(x) = \sinh x, y_2(x) = \cosh x, y(0) = 0, y'(0) = 1$$

$$[(\text{iii}). y = c_1 \sinh x + c_2 \cosh x, (\text{iv}). y = \sinh x]$$

$$21. y'' - 2y' + y = 0; y_1(x) = e^x, y_2(x) = xe^x, y(0) = 1, y'(0) = 1$$

$$[(\text{iii}). y = c_1 e^x + c_2 xe^x; (\text{iv}). y = e^x]$$

$$22. x^2 y'' - 2y = 0; y_1(x) = x^2, y_2(x) = x^{-1}, y(1) = 1, y'(1) = 1$$

$$[(\text{iii}). y = c_1 x^2 + c_2 x^{-1}, (\text{iv}). y = (2x^2 + x^{-1})/3]$$

5.8 Solution of Linear Differential Equation with Constant Coefficients

The general nth order linear differential equation with constant coefficients is given as,

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = f(x), \quad (5.8.1)$$

where all a's are constants.

Before actually solving the linear differential equations with constant coefficients, we shall examine the composition of the **complete** or **general** solution of Eqn.(5.8.1). This solution consists of two parts:

- a. **Complementary Function (or, Transient Solution), y_c**
- b. **Particular Integral/Solution (or, Steady-State Solution), y_p .**

Complementary Function (or, Transient Solution) y_c

The reduced (or homogeneous) equation obtained from Eqn (5.8.1) is ,

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = 0. \quad (5.8.2)$$

The general solution of this equation is called the **complementary function** (or **transient solution**). It is denoted by y_c .

Particular Integral/Solution (or, Steady-State Solution), y_p

If $y = y_p(x)$ is a solution of Eqn.(5.8.1) containing no arbitrary constants then this solution is called a **particular integral, or particular solution, or steady-state solution**.

Complete or General Solution.

The sum of the **complementary function** and the **particular integral**, i.e.,

$$y = y_c + y_p$$

is called the **complete** or **general solution** of Eqn.(5.8.1).

5.9 The Differential Operator D and F(D)

The symbol D is used for differential operator, $\frac{d}{dx}$. Similarly the symbols D^2, D^3, D^4 , etc are used for the differential operators, $\frac{d^2}{dx^2}, \frac{d^3}{dx^3}, \frac{d^4}{dx^4}$, etc. These operators are of great advantage in further study of linear differential equations. Therefore, the linear differential equation of n th order having constant coefficients, given by Eqn.(5.8.1) may be written in terms of the operator D as,

$$(a_n D^n + a_{n-1} D^{n-1} + \dots + a_2 D^2 + a_1 D + a_0) y = f(x) \quad (5.9.1)$$

The operator $(a_n D^n + a_{n-1} D^{n-1} + \dots + a_2 D^2 + a_1 D + a_0)$ has the form of a polynomial and may be denoted by $F(D)$. Accordingly the above differential equation can be written as,
 $F(D) y = f(x)$.

The usefulness of the symbolic operators D and $F(D)$ results from the fact that in many operations, D behaves the same way as if it were an algebraic quantity and obeys the algebraic laws. It would, however, be kept in mind that D is just an operator and not a number.

Some of the important properties of the operator D are given as under,

- a. It is distributive over the terms of a sum, i. e.,

$$D(\phi_1 + \phi_2) = D\phi_1 + D\phi_2.$$

- b. It is commutative with constants, i.e.,

$$D(c \phi) = c D(\phi).$$

- c. It obeys the index law, i.e.,

$$D^m (D^n \phi) = D^n (D^m \phi) = D^{m+n} (\phi);$$

(m, n being +ve integers).

- d. Because $D^k (a e^{bx}) = a b^k e^{bx}$, therefore,

$$F(D) (a e^{bx}) = a F(b) e^{bx}.$$

Shift Theorem for $F(D)$

It states that,

$$F(D) [e^{bx} \phi(x)] = e^{bx} [F(D + b) \phi(x)]. \quad (5.9.2)$$

Proof

This result can be verified by the use of **Leibnitz Theorem**, i. e.,

$$D^n (u v) = u (D^n v) + n (Du) (D^{n-1} v) + \frac{n(n-1)}{2!} (D^2 u) (D^{n-2} v) + \dots + (D^n u) v.$$

Therefore,

$$D^n (e^{bx} \phi) = e^{bx} [D^n \phi + nb D^{n-1} \phi + \frac{n(n-1)}{2!} b^2 D^{n-2} \phi + \dots + b^n \phi] = e^{bx} (D + b)^n \phi.$$

Thus,

$$F(D) (e^{bx} \phi) = e^{bx} \cdot F(D + b) \phi.$$

5.10 The Homogeneous Linear Differential Equation $F(D)y = 0$

The general homogeneous linear differential equation with constant coefficients is,

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = 0, \quad (5.10.1)$$

where all a 's are real constants. To find the general solution of this equation, we consider first-order, and second-order linear differential equations with constant coefficients, and then develop a general method for the solution of the n th order equation. To begin with we consider ,

$$\frac{dy}{dx} - k y = 0 \quad (5.10.2)$$

which is of **separation of variables** form. Its solution is,

$$y = ce^{kx}, \quad (5.10.3)$$

where c is an arbitrary constant. In terms of the D -operator, Eqn.(5.10.2) can be written as,

$$(D - k) y = 0,$$

its solution being given by Eqn.(5.10.3)

Next, we consider the second-order differential equation,

$$\frac{d^2 y}{dx^2} - k^2 y = 0 \quad (5.10.4)$$

In terms of the D -operator we can write it as,

$$(D^2 - k^2) y = 0$$

or,

$$[(D - k)(D + k)] y = 0; \quad (5.10.5)$$

i. e.,

$$\left(\frac{d}{dx} - k\right) y = 0, \text{ or, } \left(\frac{d}{dx} + k\right) y = 0$$

i.e.,

$$\frac{dy}{dx} - ky = 0, \text{ or, } \frac{dy}{dx} + ky = 0.$$

Both these equations being of separation of variables form, their solutions are given as,

$$y = c_1 e^{kx}, \quad y = c_2 e^{-kx},$$

where c_1 and c_2 are arbitrary constants. Therefore, the general solution of Eqn.(5.10.4) is given as,

$$y = c_1 e^{kx} + c_2 e^{-kx}. \quad (5.10.6)$$

Here, we shall verify the above. For Eqn.(5.10.6) to be a general solution of Eqn.(5.10.4), the former must satisfy the latter equation. For that we differentiate Eqn.(5.10.6) twice and substitute values of y'' , and y into the left hand side of Eqn.(5.10.4). As such, for Eqn.(5.10.4),

$$\text{LHS} = y'' - k^2 y = (ck^2 e^{kx} + ck^2 e^{-kx}) - k^2 (c_1 e^{kx} + c_2 e^{-kx}) = 0 = \text{RHS}.$$

Therefore, it is verified that Eqn.(5.10.6) is indeed the general solution of Eqn.(5.10.4),

The above solution for Eqn.(5.10.4) can also be obtained by another method, which can be generalized later. We write Eqn.(5.10.4) as,

$$F(D) y = 0,$$

where, $F(D) = D^2 - k^2$. If we make a formal substitution of $D = m$ in this equation, we obtain a polynomial in m ,

$$F(m) = m^2 - k^2.$$

If we equate this to zero, we have an algebraic equation,

$$m^2 - k^2 = 0,$$

which is called an **auxiliary** (or, **characteristic**) **equation** of the given differential equation. Its roots are,

$$m_1 = k, \quad m_2 = -k.$$

Thus, the general solution of Eqn.(5.10.4) is given as,

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x},$$

or,

$$y = c_1 e^{kx} + c_2 e^{-kx},$$

which is the same as Eqn (5.10.6).

We can generalize the above process to an n th order linear differential equation given by Eqn.(5.10.1). In terms of the D -operator, it can be written as,

$$(a_n D^n + a_{n-1} D^{n-1} + \dots + a_2 D^2 + a_1 D + a_0) y = 0. \quad (5.10.7)$$

The **auxiliary equation** of this differential equation is,

$$a_n m^n + a_{n-1} m^{n-1} + \dots + a_2 m^2 + a_1 m + a_0 = 0. \quad (5.10.8)$$

If $m_1, m_2, \dots, m_n$ are roots of this auxiliary equation, then the general solution of the differential equation (5.10.7) is,

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}. \quad (5.10.9)$$

5.11 The Nature of Roots of the Auxiliary Equation

The **auxiliary** (or **characteristic**) equation of a linear differential equation may have the following types of roots:

- a. **Real and distinct roots,**
- b. **Real and repeated roots,**
- c. **Complex conjugate (distinct/repeated) roots.**

If we consider a second-order homogeneous linear differential equation of the form,

$$y'' + a_1 y' + a_0 y = 0, \quad (5.11.1)$$

where, a_0 , and a_1 are arbitrary constants, its auxiliary equation would be,

$$m^2 + a_1 m + a_0 = 0. \quad (5.11.2)$$

The nature of the roots of Eqn.(5.11.2) depends on the sign of the discriminant $(a_1^2 - 4a_0)$. Accordingly, this equation has,

- a. Two real and distinct roots if $(a_1^2 - 4a_0) > 0$,
- b. A real and repeated (double) root if $(a_1^2 - 4a_0) = 0$,
- c. Complex conjugate roots if $(a_1^2 - 4a_0) < 0$.

We shall discuss each one of the above cases separately.

5.12 Auxiliary Equation with Distinct Real Roots

If $m_1, m_2, \dots, m_n$, are n distinct real roots of the auxiliary equation of an n th-order linear differential equation (5.10.1), then,

$$y = c_1 e^{m_1 x}, \quad y = c_2 e^{m_2 x}, \dots, y = c_n e^{m_n x},$$

are n distinct solutions of that linear differential equation. These *solutions* are *linearly independent*, if their Wronskian $W \neq 0$. Thus the general solution of the differential equation is,

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x} \quad (5.12.1)$$

Example 5.12.1

Solve the differential equation:

$$y'' - 5y' + 6y = 0.$$

Solution

The given equation can be written with the help of D-operator as:

$$(D^2 - 5D + 6) y = 0.$$

Its auxiliary equation is

$$m^2 - 5m + 6 = 0, \text{ or } (m - 2)(m - 3) = 0.$$

Therefore, the auxiliary equation has the real distinct roots: $m_1 = 2$, and $m_2 = 3$. Thus, the general solution of the given differential equation is given as,

$$y = c_1 e^{2x} + c_2 e^{3x}.$$

Example 5.12.2

Solve the differential equation

$$y'' + y' - 12y = 0.$$

Solution

The given equation can be written as, $(D^2 + D - 12)y = 0$, so that its auxiliary equation is,

$$m^2 + m - 12 = 0.$$

Its roots are: $m_1 = 3$, $m_2 = -4$. Thus, the general solution of the given differential equation is,

$$y = c_1 e^{3x} + c_2 e^{-4x}.$$

EXERCISE-5.3

(Auxiliary Equation With Real and Distinct Roots)

Note: Answers are given against the respective questions within square brackets.

Solve the differential equations,

1. $y'' + 3y' - 54y = 0;$ $[y = c_1 e^{6x} + c_2 e^{-9x}]$
2. $y'' + m^2 y = 0;$ $[y = c_1 e^{mx} + c_2 e^{-mx}]$
3. $y'' + 4y' = 0;$ $[y = c_1 + c_2 e^{-4x}]$
4. $y'' - 3y' + 2y = 0;$ $[y = c_1 e^x + c_2 e^{2x}]$
5. $y'' + 4y' + 3y = 0;$ $[y = c_1 e^{-x} + c_2 e^{-3x}]$
6. $y'' + 7y' + 12y = 0;$ $[y = c_1 e^{-3x} + c_2 e^{-4x}]$
7. $y'' + y' - 6y = 0;$ $[y = c_1 e^{-3x} + c_2 e^{2x}]$
8. $y'' - 4y' + 3y = 0;$ $[y = c_1 e^x + c_2 e^{3x}]$
9. $y''' - 2y'' - y' + 2y = 0;$ $[y = c_1 e^{-x} + c_2 e^x + c_3 e^{2x}]$
10. $y''' - y' = 0;$ $[y = c_1 + c_2 e^{-x} + c_3 e^x]$
11. $y''' - 6y'' + 11y' - 6y = 0;$ $[y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}]$
12. $y''' - 4y'' + y' + 6y = 0;$ $[y = c_1 e^{-x} + c_2 e^{2x} + c_3 e^{3x}]$
13. $y''' - 3y'' - y' + 3y = 0;$ $[y = c_1 e^x + c_2 e^{-x} + c_3 e^{3x}]$
14. $y''' - 6y'' + 5y' + 12y = 0;$ $[y = c_1 e^{-x} + c_2 e^{3x} + c_3 e^{4x}]$

$$15. \quad y'''' + 6 y''' + 11 y'' + 6 y' = 0; \quad [y = c_1 + c_2 e^{-x} + c_3 e^{-2x} + c_4 e^{-3x}]$$

$$16. \quad \ddot{x} - 3\dot{x} + 2(x-1) = 0; \quad (\text{Hint : put } x-1=y) \quad [x = 1 + c_1 e^t + c_2 e^{2t}]$$

$$17. \quad 2\ddot{x} + 5\dot{x} - 12x = 0; \quad [x = c_1 e^{(3/2)t} + c_2 e^{-4t}]$$

$$18. \quad 9\ddot{x} + 18\dot{x} - 16x = 0; \quad [x = c_1 e^{(2/3)t} + c_2 e^{(-8/3)t}]$$

Solve the initial value problems:

$$19. \quad y'' + 8y' - 9y = 0, y(1) = 1, y'(1) = 0; \quad [y = e^{-9(x-1)}/10 + 9e^{x-1}/10]$$

$$20. \quad y'' + 3y' + 2y = 0, y(0) = 2, y'(0) = 0; \quad [y = 4e^{-x} - 2e^{-2x}]$$

$$21. \quad y'' - y' - 6y = 0, y(0) = 1, y'(1) = 0; \quad [y = (2e^{3x} - 3e^{5-2x})/(2 + 3e^5)]$$

$$22. \quad y''' - 6y'' + 11y' - 6y = 0, y(0) = 0, y'(0) = 0, y''(0) = 2; \quad [y = e^x - 2e^{2x} + e^{3x}]$$

5.13 Auxiliary Equation with Repeated Real Roots

When two or more roots of the auxiliary equation of an n th-order linear differential equation are equal (or coincident), the procedure discussed in the preceding section does not apply, as may be seen from the following discussion.

Suppose an n th-order linear differential equation has two equal (or coincident) roots, say, $m_1 = m_2$. Then in view of our earlier discussion, the solution (5.10.9) would be,

$$y = (c_1 + c_2)e^{m_1x} + c_3 e^{m_3x} + \dots + c_n e^{m_nx}. \quad (5.13.1)$$

But c_1 and c_2 are arbitrary constants, and their sum can be replaced by a new arbitrary constants, say $c_1 + c_2 = c$. Thus the solution no longer contains n arbitrary constants (their number being reduced to $n-1$), which is an essential requirement for it to become the general solution. We may overcome this difficulty and find the requisite general solution by different methods. Here, we illustrate two plausible methods as discussed below:

First Method

To tackle the problem of finding the general solution of a differential equation whose auxiliary (or, characteristic) equation has repeated roots, we have the following theorem:

Theorem

If we know one solution, say, $y = y_1$ of an n th-order linear differential equation $F(D)y = 0$, then the substitution $y = y_1v$, will transform the given equation into an equation of the $(n-1)$ st order in v' , i.e., dv/dx .

(This theorem is valid for constant or variable coefficients and is also valid for equation $F(D)y = f(x)$).

To illustrate the above theorem, we find the general solution of the linear differential equation,

$$y'' - 6y' + 9y = 0. \quad (5.13.2)$$

This equation can be written as,

$$(D^2 - 6D + 9)y = 0. \quad (5.13.3)$$

The auxiliary equation for this differential equation is,

$$m^2 - 6m + 9 = 0, \text{ or, } (m - 3)^2 = 0,$$

which has the repeated roots: $m = 3, 3$. Thus $y = e^{3x}$ is one solution of Eqn.(5.13.2). Therefore, according to the above theorem, we let $y = ve^{3x}$, so that,

$$Dy = v' e^{3x} + 3v e^{3x}, \quad D^2y = v'' e^{3x} + 6v' e^{3x} + 9v e^{3x}.$$

On substituting these values into Eqn.(5.13.3), we get on simplifying,

$$v'' e^{3x} = 0.$$

Here $e^{3x} \neq 0$. Thus $v'' = 0$, which gives $v = c_0 + c_1 x$. As such we get the general solution of the differential equation (5.13.2) as,

$$y = (c_0 + c_1 x) e^{3x}. \quad (5.13.4)$$

Similarly if we have a differential equation whose auxiliary equation, say, $(m - 2)^3 = 0$, has three repeated roots, $m = 2, 2, 2$, then the general solution is,

$$y = (c_0 + c_1 x + c_2 x^2) e^{2x}. \quad (5.13.5)$$

If the auxiliary equation has k repeated roots, say, $m = r, r, r, \dots$ (k times), then the general solution of the respective differential equation is ,

$$y = (c_0 + c_1 x + c_2 x^2 + \dots + c_{k-2} x^{k-2} + c_{k-1} x^{k-1}) e^{rx}. \quad (5.13.6)$$

Second Method.

Suppose the auxiliary equation of a second-order linear differential equation has two repeated roots, say, m, m . Then it can be written as,

$$(D - m)^2 y = 0; \quad (5.13.7)$$

or,

$$(D - m)(D - m)y = 0.$$

We let,

$$(D - m)y = v. \quad (5.13.8)$$

Thus, Eqn (5.13.7) can be written as,

$$(D - m)v = 0;$$

or,

$$\frac{dv}{dx} - mv = 0.$$

This equation being of “separation of variables” form, its solution is given as,

$$v = c_1 e^{mx}.$$

Substituting this value of v in Eqn (5.13.8) we obtain,

$$(D - m)y = c_1 e^{mx};$$

or,

$$\frac{dy}{dx} - my = c_1 e^{mx}. \quad (5.13.9)$$

It is a first order linear differential equation (of the form $y' + Py = Q$), with an *integrating factor* e^{-mx} . It can be solved by technique 6 (see sec. 3.9). Its solution is given as,

$$y = e^{mx} \left[\int e^{-mx} (C_1 e^{mx}) dx + C_0 \right]$$

or,

$$y = [C_0 + C_1 x] e^{mx},$$

which is the required solution of Eqn.(5.13.7).

Similarly, if a third order linear differential equation has three repeated roots, say, m, m, m , then it can be written as,

$$(D - m)^3 y = 0.$$

Proceeding in the manner as mentioned above (in the case of a second order differential equation, its auxiliary equation having two repeated roots), we obtain the general solution as,

$$y = (C_0 + C_1 x + C_2 x^2) e^{mx}. \quad (5.13.10)$$

If the auxiliary equation of a k th order linear differential equation has k repeated roots, say, $m, m, m, \dots$ (k times), the general solution of that equation would be,

$$y = (C_0 + C_1 x + C_2 x^2 + \dots + C_{k-1} x^{k-1}) e^{mx}. \quad (5.13.11)$$

Examples 5.13.1

Solve the differential equation,

$$y''' - 5y'' + 7y' - 3y = 0. \quad (5.13.12)$$

Solution

The given equation can be written as,

$$(D^3 - 5D^2 + 7D - 3)y = 0,$$

whose auxiliary equation is,

$$m^3 - 5m^2 + 7m - 3 = 0.$$

The roots of this auxiliary equation are, $m = 1, 1, 3$ (one distinct and two repeated roots). Therefore, the general solution of the given differential equation is,

$$y = (C_0 + C_1 x) e^x + C_2 e^{3x}. \quad (5.13.13)$$

EXERCISE-5.4

(Auxiliary Equation with Repeated Real Roots)

Note: Answers are given against the respective questions within square brackets

Solve the differential equations:

1. $y'' + 4y' + 4y = 0;$ $[y = (c_0 + c_1 x) e^{-2x}]$
2. $y'' + 6y' + 9y = 0;$ $[y = (c_0 + c_1 x) e^{-3x}]$
3. $4y'' + 4y' + y = 0;$ $[y = (c_0 + c_1 x) e^{-x/2}]$
4. $y''' - 6y'' + 12y' - 8y = 0;$ $[y = (c_0 + c_1 x + c_2 x^2) e^{2x}]$
5. $y''' - y'' - y' + y = 0;$ $[y = (c e^{-x} + (c_0 + c_1 x) e^x)]$
6. $y''' - 3y'' + 3y' - y = 0;$ $[y = (c_0 + c_1 x + c_2 x^2) e^x]$
7. $y'''' - 5y''' + 6y'' + 4y' - 8y = 0;$ $[y = (c_0 + c_1 x + c_2 x^2) e^{2x} + c_3 e^{-x}]$
8. $y'''' - y''' - 9y'' - 11y' - 4y = 0;$ $[y = (c_0 + c_1 x + c_2 x^2) e^{-x} + c_3 e^{4x}]$
9. $y'''' - 6y''' + 9y'' = 0;$ $[c_0 + c_1 x + (c_2 + c_3 x) e^{3x}]$
10. $y'''' - y''' - 3y'' + y' + 2y = 0;$ $[y = c e^x + (c_0 + c_1 x) e^{-x} + c_2 e^{2x}]$
11. $y^{(v)} - 2y'''' + y''' = 0;$ $[y = c_0 + c_1 x + c_2 x^2 + (c_3 + c_4 x) e^x]$
12. $\ddot{x} + 4\dot{x} + 4x = 0$ $[x = (c_0 + c_1 t) e^{-2t}]$

Solve the initial value problem:

13. $y'' + 6y' + 9y = 0, \quad y(0) = 2, y'(0) = -3;$ $[y = (2 + 3x) e^{-3x}]$
14. $y'' - 6y' + 9y = 0, \quad y(0) = 0, y'(0) = 2;$ $[y = 2x e^{3x}]$
15. $y'' - 2y' + y = 0, \quad y(0) = 5, y'(0) = -9;$ $[y = (5 - 14x) e^x]$
16. $y'' + 2y' + y = 0, \quad y(0) = 1, y'(0) = -1;$ $[y = e^{-x}]$
17. $y''' - y'' = 0, \quad y(0) = 2, y'(0) = 3, y''(0) = 2;$ $[y = x + 2e^x]$
18. $y''' + y'' = 0, \quad y(0) = 4, y'(0) = -2, y''(0) = 4;$ $[y = 2x + 4e^{-x}]$
19. $y''' - 4y'' + 4y' = 0, \quad y(0) = 1, y'(0) = 2, y''(0) = 8;$ $[y = 1 + 2x e^{2x}]$
20. $y''' - y'' - y' + y = 0, \quad y(0) = 0, y'(0) = 0, y''(0) = 4;$ $[y = e^{-x} + (2x - 1) e^x]$

$$21. \quad y''' - 6y'' + 9y' = 0, \quad y(0) = 0, \quad y'(0) = 2, \quad y''(0) = -6; \quad [y = -2 + e^{3x}(2 - 4x)]$$

5.14 Auxiliary Equation with Complex Roots.

If the auxiliary equation of a differential equation has complex roots, they always occur in conjugate pairs. Thus if $(p + iq)$ is one root, the other root must be $(p - iq)$, where p, q , are real and $q \neq 0$. Assume now that $p + iq$ and $p - iq$ are the complex roots of the auxiliary equation of a second order linear differential equation. Since they are **distinct roots**, the general solution is given by (like in the case of real distinct roots, see sec 5.12),

$$y = c_1 e^{(p+iq)x} + c_2 e^{(p-iq)x} = e^{px} [c_1 e^{iqx} + c_2 e^{-iqx}].$$

But we know that, $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$ (*Euler's formula*). Therefore,

$$\begin{aligned} y &= e^{px} [c_1(\cos qx + i \sin qx) + c_2(\cos qx - i \sin qx)] \\ &= e^{px} [(c_1 + c_2) \cos qx + (i c_1 - i c_2) \sin qx]. \end{aligned}$$

Thus,

$$y = e^{px} (A \cos qx + B \sin qx),$$

where, $A = c_1 + c_2$, and $B = i(c_1 - c_2)$ are arbitrary constants.

If a pair of complex roots, $p \pm iq$, occurs twice, the corresponding general solution is,

$$y = (c_1 + c_2 x) e^{(p+iq)x} + (c_3 + c_4 x) e^{(p-iq)x}.$$

It reduces to,

$$y = e^{px} [(A_0 + A_1 x) \cos qx + (B_0 + B_1 x) \sin qx],$$

where, $A_0 = c_1 + c_3$, $A_1 = c_2 + c_4$, $B_0 = i(c_1 - c_3)$, $B_1 = i(c_2 - c_4)$ are arbitrary constants.

Repeated Complex Roots.

If the pair of complex roots, $p + iq$ occurs k times, the corresponding general solution of a differential equation can be written as,

$$y = e^{px} [(A_0 + A_1 x + A_2 x^2 + \dots + A_{k-1} x^{k-1}) \cos qx + (B_0 + B_1 x + B_2 x^2 + \dots + B_{k-1} x^{k-1}) \sin qx]$$

Example 5.14.1.

Solve the differential equation,

$$y'' + k^2 y = 0.$$

Solution

The given equation can be written as,

$$(D^2 + k^2) y = 0,$$

which has the auxiliary equation,

$$m^2 + k^2 = 0.$$

The roots of this equation are, $m = \pm i k$, i.e., $m = 0 \pm i k$. Therefore, the general solution of the given differential equation is, $y = e^{0x} (A \cos kx + B \sin kx)$, i.e.,

$$y = A \cos kx + B \sin kx$$

Examples 5.14.2.

Solve the differential equation,

$$y''' - y'' + y' - y = 0..$$

Solution

The given equation can be written as,

$$(D^3 - D^2 + D - 1) y = 0.$$

Thus the auxiliary equation is,

$$m^3 - m^2 + m - 1 = 0, \text{ or, } (m - 1)(m^2 + 1) = 0.$$

Its roots are, $m = 1, \pm i$, i.e., $m = 1, 0 \pm i$. Therefore, the general solution of the given differential equation is,

$$y = c e^x + (A \cos x + B \sin x).$$

EXERCISE-5.5

(Auxiliary Equation with Complex Roots)

Note: Answers are given against the respective questions within square brackets.

Solve the differential equation:

1. $y'' + y = 0$ [$y = A \cos x + B \sin x$]
2. $y'' + 2y' + 2y = 0$; [$y = e^{-x} (A \cos x + B \sin x)$]
3. $y'' - 4y' + 8y = 0$; [$y = e^{2x} (A \cos 2x + B \sin 2x)$]
4. $8y'' + 4y' + y = 0$; [$y = e^{-x/4} \{A \cos (x/4) + B \sin (x/4)\}$]
5. $y'' - 2y' + 5y = 0$; [$y = e^x (A \cos 2x + B \sin 2x)$]
6. $y'' + 9y = 0$; [$y = A \cos 3x + B \sin 3x$]
7. $y'' - 6y' + 25y = 0$; [$y = e^{3x} (A \cos 4x + B \sin 4x)$]
8. $y'' - 4y' + 5y = 0$; [$y = e^{2x} (A \cos x + B \sin x)$]
9. $\frac{d^2s}{dt^2} + 4\frac{ds}{dt} + 13s = 0$; [$s = e^{-2t} (A \cos 3t + B \sin 3t)$]
10. $y''' - 4y'' + 5y' = 0$; [$y = C + e^{2x} (A \cos x + B \sin x)$]
11. $y''' - 3y'' + 9y' + 13y = 0$; [$y = C e^{-x} + e^{2x} (A \cos 3x + B \sin 3x)$]
12. $y''' - 3y'' + 4y' - 2y = 0$; [$y = e^x (C + A \cos x + B \sin x)$]
13. $y''' + 8y = 0$; [$y = A e^{-2x} + e^x \cos (x\sqrt{3} - \alpha)$], $\alpha = \tan^{-1}(B/A)$]
14. $y'''' + 4y = 0$; [$y = e^x (A \cos x + B \sin x) + e^{-x} (C \cos x + D \sin x)$]
15. $y'''' + 18y'' + 81y = 0$; [$y = (c_0 + c_1x) \cos 3x + (c_2 + c_3x) \sin 3x$]
16. $y'''' + 8y'' + 16y = 0$; [$y = (c_0 + c_1x) \cos 2x + (c_2 + c_3x) \sin 2x$]
17. $y'''' + 13y'' + 36y = 0$; [$y = A \cos 2x + B \sin 2x + C \cos 3x + D \sin 3x$]
18. $y^{(6)} + 6y'''' + 9y'' = 0$; [$y = c_0 + c_1x + (c_2 + c_3x) \cos \sqrt{3}x + (c_4 + c_5x) \sin \sqrt{3}x$]

Solve the initial value problems,

19. $y'' + y = 0$, $y(0) = y_0$, $y'(0) = 0$; [$y = y_0 \cos x$]

20. $y'' + y = 0$, $y(\pi) = 1$, $y'(\pi) = -1$; $[y = \sin x - \cos x]$
21. $y'' + 2y' + 2y = 0$, $y(0) = 0$, $y'(0) = 1$; $[y = e^{-x} \sin x]$
22. $y'' + 9y = 0$, $y(\pi/6) = 2$, $y'(\pi/6) = 0$; $[y = 2 \sin 3x]$
23. $y'' + 4y = 0$, $y(1) = 4$, $y'(1) = 0$; $[y = 4 \cos 2(x-1)]$
24. $y'' + 6y' + 13y = 0$, $y(0) = 3$, $y'(0) = -1$; $[y = e^{-3x} (3 \cos 2x + 4 \sin 2x)]$
25. $y'' + y = 0$, $y(\pi/3) = 2$, $y'(\pi/3) = -4$; $[y = (1 + 2\sqrt{3}) \cos x - (2 - \sqrt{3}) \sin x]$
26. $y'' - 4y' + 20y = 0$, $y(0) = 0$, $y'(0) = 8$; $[y = 2e^{2x} \sin 4x]$
27. $y'' - 2y' + 5y = 0$, $y(0) = 2$, $y'(0) = 4$; $[y = e^x (2 \cos 2x + \sin 2x)]$
28. $y'' - 4y' + 20y = 0$, $y(\pi/2) = 0$, $y'(\pi/2) = 1$; $[y = (1/4)e^{2x-\pi} \sin 4x]$
29. $I \frac{d^2\theta}{dt^2} + g\theta = 0$; given that $\theta = \alpha$, and $d\theta/dt = 0$ at $t = 0$; $[\theta = \alpha \cos t \sqrt{g/l}]$
30. $y''' + 5y'' + 17y' + 13y = 0$, $y(0) = 0$, $y'(0) = 1$, $y''(0) = 6$; $[y = e^{-x} - e^{-2x} \cos 3x]$
31. $y''' + y'' + 4y' + 4y = 0$, $y(0) = 0$, $y'(0) = -1$, $y''(0) = 5$; $[y = e^{-x} - \cos 2x]$
32. $y''' - 2y'' + 2y' = 0$, $y(0) = 1$, $y'(0) = 1$, $y''(0) = 2$; $[y = 1 + e^x \sin x]$
33. $y''' - 2y' + 4y = 0$, $y(0) = -2$, $y'(0) = 8$, $y''(0) = 0$; $[y = 4e^x \sin x - 2e^{-2x}]$

CHAPTER – 6

*NON-HOMOGENEOUS LINEAR DIFFERENTIAL EQUATIONS***6.1 General**

A differential equation of the form,

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + a_{n-2} y^{(n-2)} + \dots + a_1 y' + a_0 y = f(x), \quad (6.1.1)$$

with $f(x) \neq 0$, is called a ***non-homogeneous linear differential equation of n th order***. Using the D-operator it can be written as,

$$(a_n D^n + a_{n-1} D^{n-1} + \dots + a_1 D + a_0) y = f(x),$$

i. e.,

$$F(D) y = f(x). \quad (6.1.2)$$

Its **solution** consists of two parts:

- a. Complementary Function (or Transient Solution), y_c**
- b. Particular Integral/Solution (or Steady-State Solution), y_p .**

The complementary function, y_c , is the solution of homogeneous differential equation, $F(D)y = 0$, corresponding to the above mentioned differential equation. It has been discussed in Chapter 5. The particular integral, y_p , is the solution obtained from the above non-homogeneous differential equation (6.1.2) as, $y_p = \frac{1}{F(D)} \{f(x)\}$, where $\frac{1}{F(D)}$, is the inverse differential operator (discussed below).

Important Formulae to Evaluate y_p

We shall develop the following formulae to find the ***particular integral/solution (steady-state solution)***, y_p :

- a. $\frac{1}{F(D)} \{a e^{bx}\} = \frac{a e^{bx}}{F(b)}$, where, $F(b) \neq 0$,
- b. $\frac{1}{F(D)} \{a e^{bx}\} = \frac{1}{(D-b)^r \phi(D)} \{a e^{bx}\} = \frac{a e^{bx}}{\phi(b)} \frac{x^r}{r!}$, (when $F(b) = 0$ in the above formula).
- c. $\frac{1}{F(D)} \{a e^{bx} f(x)\} = a e^{bx} \frac{1}{F(D+b)} \{f(x)\}$,
- d. $\frac{1}{F(D)} \{\text{Polynomial}\} = \frac{1}{k} \left[\frac{1}{1-F_1(D)} \right] \{\text{Polynomial}\} = \frac{1}{k} [1 - F_1(D)]^{-1} \{\text{Polynomial}\}$
 $= \frac{1}{k} [1 + c_1 D + c_2 D^2 + c_3 D^3 + \dots] \{\text{Polynomial}\}$,
 (where $k, c_1, c_2, c_3, \dots$, are constants)

OR

$$\begin{aligned} \frac{1}{F(D)} \{\text{Polynomial}\} &= \frac{1}{k} \left[\frac{1}{1+F_1(D)} \right] \{\text{Polynomial}\} = \frac{1}{k} [1 + F_1(D)]^{-1} \{\text{Polynomial}\} \\ &= \frac{1}{k} [1 - c_1 D + c_2 D^2 - c_3 D^3 + \dots] \{\text{Polynomial}\} \\ &\text{(where } k, c_1, c_2, c_3, \dots, \text{ are constants).} \end{aligned}$$

6.2 The Algebra of Differential Operators

Let,

$$L_1 = F(D) = a_n D^n + a_{n-1} D^{n-1} + \dots + a_0, \text{ and}$$

$$L_2 = f(D) = b_n D^n + b_{n-1} D^{n-1} + \dots + b_0,$$

be two differential operators, where $D = d/dx$, $D^2 = d^2/dx^2$, $\dots$, $D^n = d^n/dx^n$; and the coefficients may depend on x . It may be seen that the usual fundamental laws of algebra hold for the operators $L_1, L_2, \dots$, etc. That is,

- a. $(L_1 + L_2) y = L_1 y + L_2 y$,
- b. $(kL_1) y = k(L_1) y$, and $[g(x) L_1] y = g(x) [L_1 y]$
- c. $L_1 + L_2 = L_2 + L_1$,
- d. $L_1 + (L_2 + L_3) = (L_1 + L_2) + L_3$,
- e. $k(L_1 + L_2) = k L_1 + k L_2$, and $g(x) [L_1 + L_2] = g L_1 + g L_2$,
- f. $(L_1 L_2) y = L_1 (L_2 y) = L_2 (L_1 y)$,
- g. $L_1 L_2 = L_2 L_1$,
- h. $L_1 (L_2 L_3) = (L_1 L_2) L_3$,
- j. $L_1 (L_2 + L_3) = L_1 L_2 + L_1 L_3$.